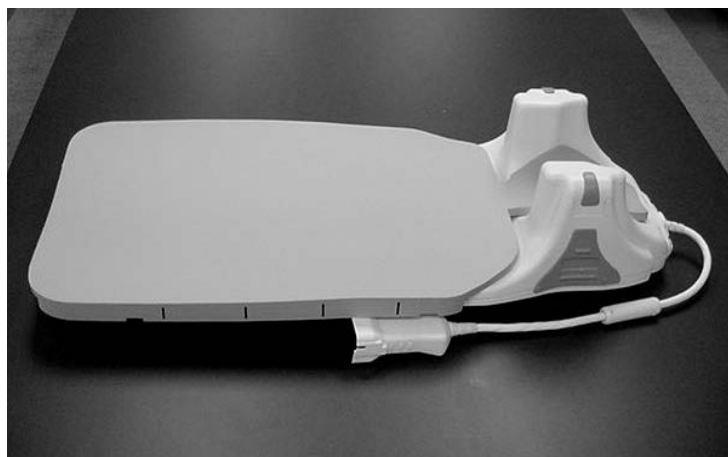


3.0T HD CTL Array

8-Channel

Operator Manual

2422232-1EN



Operator Manual for the 3.0T HD
CTL Array
2422232-1EN
Revision 04 (2023-05)

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Revision History

Table i: Revision History

Revision	Date	Author	Reason for Change
01	2013-02	D. Seaman	First Release
02	2014-02	D. Seaman	All - updated revision, revision date, copyright date Page i - added "USA" to manufacturer address
03	2018-05	D. Seaman	• Throughout - removed "Healthcare" from GE name except where required, updated revision, revision date and date format, changed caution symbol background color • Title Page - removed "GE Healthcare Doing Business as..." statement, removed GE Healthcare from upper left corner, added CE symbol • Page i - added revision history, the year of CE marking sentence • Page ii - added Environmental Requirements section • Page iii - added coil storage section, added contact Information table and in table moved the GE website and added USA to manufacturer address • Page iv - added eIFU section • Page 1-4 - updated upper do not loop symbol, updated both do not loop symbol definitions • Page 1-5 - added 2nd receive coil symbol, added caution symbol, added do not insert with this side up symbol, updated fragile and keep dry symbols • Page 1-6 - replaced transportation label with individual temperature, humidity and atmospheric pressure symbols • Page 2-1 - added Intended use statement • Page 2-5 - added do not loop label found on old revision coils, added do not insert connector plug warning • Pages 3-3 to 3-8 - updated text in images for translation • Page 3-9 - anonymized image for step 2 • Chapter 4 - removed USA phone number from Quality Assurance and Accessories sections

Table i: Revision History (Continued)

Revision	Date	Author	Reason for Change
04	2023-05	R. Goe	-Throughout manual, updated all font to Calibri, all footers to Rev. 4, 2023-05, 2023 GE HealthCare, changed GE Healthcare to GE HealthCare except for “Manufacturer Address”. -Title Page – Updated Revision to 04 (2023-05), 2023 GE HealthCare, removed CE number, update to GE Logo. -Page iii – Updated from MDD to Medical Device Regulation, removed CE number, removed GE HealthCare Coils address, added trash can without black bar. -Page iv – Inserted “Receiving Your Coil” caution and Rights and License and Trademarks”, added table “Headings” to all tables. -Page v – Inserted “Radio Frequency Energy Information” page. -Page vii – Updated “Electronic Instructions for Use”. -Page 1-3 – Inserted “Clinical Benefits of MR Coils” paragraph. -Page 1-4 - Inserted 2nd trash can without black bar. -Page 1-5 – Added third “Receive Only” symbol with “R”, added new symbols for the “Caution” and “Warning” symbols, added FRU, REV, and FIELD symbols and definitions. -Page 1-7 – Inserted “eIFU, Radio Frequency Energy, MD, Swiss Representative, European Representative and Important symbols and definitions. -Page 2-1 – Added “Introduction to Safety” header, added “Contraindications”, updated heading to “Intended Use/ Indications for Use”. -Page 2-2, 2-3 – Removed two warnings and replace with one stating “RF can cause localized heating...”. -Page 2-5 – Updated “Modifications” warning and “This serial number...”. -Page 2-6 – Updated “Emergency Procedures” section. -Page 3-1 – Moved “Coil Installation” from section 1. -Page 4-1 – Inserted three bullet points into “Coil Care” section. -Page 4-2, 4-3, 4-4 – Updated “Cleaning and Disinfecting” section. -Glossary 1, 2 – Inserted “Glossary” at end of manual.

Medical Device Regulation

This product conforms with the requirements of the Medical Device Regulation 2017/745 when they bear the following CE Mark of conformity:



The year of CE Marking is 2005.

EU Authorized Representative:



**GE Medical Systems S.C.S.
Quality Assurance Manager**
283 Rue de la Minière
78530 BUC France
Telephone: +33 1 30 70 40 40

Swiss Authorized Representative



GE Medical Systems (Schweiz) AG
Europa-Strasse 31
8152 Glattbrugg
Switzerland



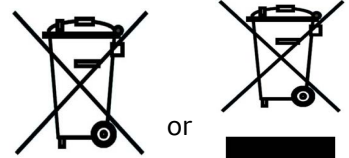
CAUTION:

United States federal law restricts the sale, distribution and use of this device to, or on the order of, a physician.



CAUTION:

This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.



Receiving Your Coil



CAUTION:

All packages should be examined at time of delivery. Please look for damaged packaging and inspect each coil component, cable and accessory to ensure there is no equipment damage from shipping. Visually check the coil for tears or damage. Ensure no internal conductors are exposed. If any coil component, cable, or accessory is suspected of not being in good condition, please consult a GE HealthCare representative before use.

Rights

All rights reserved. No part of this publication may be reproduced, transmitted, transcribed, stored in a retrieval system, or translated into any language in any means without the written permission of GE HealthCare.

License and Trademarks

Proper performance of this coil is guaranteed only while the coil is being used on the MR system (hardware/software level) specified at the time of purchase. Upgrades or other modifications to the system software and /or hardware may affect compatibility. Prior to upgrading your MR system, please contact your GE HealthCare representative to discuss coil compatibility issues. Failure to do so may void your warranty.

GE is a trademark of General Electric Company used under trademark license.

Environmental Requirements

Table ii: Environmental Requirements

Environmental Factor	Transport Conditions	Operating and Storage Conditions
Barometric pressure	645 mm Hg to 795 mm Hg (860 hPa to 1060 hPa)	As established by the GE HealthCare recommended MR suite conditions.
Humidity	20% to 95% non-condensing	
Temperature	-15°C to 55°C (5°F to 131°F)	

Radio Frequency Energy Information

This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause radio frequency interference with other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, the GE HealthCare MR Systems comply with emissions limits for (Group 2, Class A) Medical Devices as stated in EN60601-1-2. However, there is no guarantee that interference will not occur in a particular installation.



If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):

- Re-orient or relocate the affected device(s);
- Increase the separation between the equipment and the affected device;
- Power the equipment from a source different from that of the affected device and/ or
- Consult the point of purchase or service representative for further suggestions

The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.

Do not use devices, which intentionally transmit RF Signals (cellular phones, transceivers, or radio- controlled products) in the vicinity of this equipment as they may cause performance outside the published specifications. Keep the power to these types of devices turned off when near this equipment.

The medical staff in charge of this equipment is required to instruct technicians, patients, and other people who may be around this equipment to fully comply with the above requirement.

Immunity/Emissions Exceptions: Note the exceptions from the EMC test results. Check with the business EMC engineer for this information.

Coil Storage

When the coil is not in transport and is not being used, store the coil in an air conditioned area that has environmental conditions that fall within the established GE HealthCare recommended MR suite conditions.

The coil must be stored in a storage space that exceeds the dimensions of the coil.

Contact Information

Table iii: Contact Information

GE HealthCare Americas	
USA	800-558-5102
Canada	800-668-0732
GE HealthCare Global	
China	86-21-62192228
Taiwan	886-2-2505-7900
Singapore	65-291-8528
Australia	61-2-9975-5501
Japan	81-120-48-2630
Korea	82-31-740-6119
India	91-80-845-2923
Europe	33-1-41-19-76-76
GE HealthCare Website	www.gehealthcare.com
GE HealthCare Manufacturer Information	
Manufacturer Address	GE Healthcare Coils 1515 Danner Drive Aurora, Ohio 44202-9273 USA
Manufacturer Telephone Number	1-330-562-1000

Electronic Instructions for Use



Your MR device is delivered with the electronic Instructions For Use (eIFU) logo. The electronic Instructions For Use (eIFU) symbol means that the operator manual can be viewed from a PC. Use these steps to view your eIFU manual from a personal computer.

1. Insert the multi-language DVD into your personal computer's CD-ROM drive. The minimum PC requirements include the following:
 - Pentium 90 MHz
 - Windows 95
 - 32 MB of RAM
 - CD-ROM drive
 - Adobe Acrobat Reader 4.05 or higher. If your personal computer does not have Acrobat Reader you can download it for free from the Adobe website (www.adobe.com).
2. Typically, the DVD will auto start. If not, Double-click the My Computer icon.
3. Double-click the CD icon.
4. Double-click the desired language to open the pdf file.

Refer to e-User Leaflet 5452943-199, 5851600-199 or to the e-User Leaflet that came with your MR system for further instructions.

A paper copy of Instructions for Use can be ordered at no additional cost. Please send a request to your Sales or Service representative. They will transfer your request to CEMEURDIST@med.ge.com. In application of EU Commission Regulations on electronic instructions for use of medical devices, in the European Union, your request should be processed within 7 days.

To obtain the latest version of the Instructions for Use, go to:

<https://www.gehealthcare.com/documentationlibrary> - Click on the “Enter Customer Documentation Portal” button.

From the Customer Documentation Portal, enter 2422232-399 in the search field, and launch the search. The search will return a list of the available manuals. Scroll through the available language translations of the manual and download your preferred translation.

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Chapter 1

Introduction and Coil Specifications

Introduction

This manual describes the safety precautions, features, use and care of the 3.0T, 8-Channel HD CTL Arrays. The following CTL Arrays are compatible with the systems listed:

- The Premier III Phased Array CTL Spine Coil (Receive Only) - 3.0T GE HealthCare MRI Systems with 3.0T Legacy connector
- The 3.0T HD CTL Array - 3.0T GE HealthCare MRI Systems with HD connector

Please review this manual thoroughly before using the device for imaging. Refer to the MR system manual in conjunction with this manual. For questions or comments on this manual, or assistance with the use of this product, please contact a GE HealthCare representative.

This equipment is for use by qualified personnel only.

NOTE: This manual should be kept with the equipment and available at all times. It is important to periodically review the procedures and safety precautions. **All users must read and understand the contents of this manual before attempting to use this product.**

NOTE: Throughout this manual the Premier III Phased Array CTL Spine Coil (Receive Only) will either be referred to by its full name or as the 3.0T HD CTL Array. The 3.0T HD CTL Array with HD connectors will be referred to by its official name of 3.0T HD CTL Array or as the 3.0T HD CTL Array, A/B Port. The two coils will collectively be referred to as the 3.0T HD CTL Arrays.

Premier III Phased Array CTL Spine Coil (Receive Only) Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-1 Premier III Phased Array CTL Coil (Receive Only) Shipping Contents:

Part Name	Part Number	Qty.
Premier III Phased Array CTL Coil (Receive Only)	2413107	1
Service Manual	2413107-3	1
Operator Manual Packet	2422232-500	1
Wedge/Ear Pad	2414423	2
Patient Comfort Pad	2413401	1
Disposable Ear Pad Covers	U1-150536	500

3.0T HD CTL Array, A/B Port Shipping Contents

NOTE: Shipping contents may be packaged in more than one carton.

Table 1-2 3.0T HD CTL Array, A/B Port Shipping Contents:

Part Name	Part Number	Qty.
3.0T HD CTL Array, A/B Port	2415542	1
Operator Manual Packet	2422232-500	1
Wedge/Ear Pad	2414423	2
Patient Comfort Pad	2413401	1
Disposable Ear Pad Covers	U1-150536	500

Use of Symbols

Coil users should understand the following symbols used for caution and warning explanations:



Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.

CAUTION:

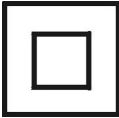


Indicates a hazard that may cause severe personal injury, death, or substantial property damage if instructions are not followed.

WARNING:

Coil Classification

Table 1-3 Classification of 3.0T CTL Coil:

Symbol	Classification
	Class II equipment
	Type BF applied part

- Suitable for continuous operation.
- The IP Rating for the coil is IPX0. Ordinary equipment. Not rated against water ingress protection.
- Non-sterile device.
- Equipment not suitable for use in the presence of a flammable anesthetic mixture with air or with oxygen or nitrous oxide.
- The voltage rating of the coil is 10VDC maximum.

All surface areas of the coil, except the bottom surface that sits on the patient table, are considered applied parts and may come into direct contact with the patient. Pads and/or thermal-resistant materials should be used to prevent the coil surfaces from touching the patient.

Pads and/or thermal-resistant material must be used to ensure the patient cannot touch the coil cable and connectors.

Clinical Benefit of MR Coils

The clinical benefit of an MR Coil is to help the healthcare professional provide accurate diagnostic information that enhances the diagnostic and treatment care pathways of the patient for a variety of diseases and conditions.

Symbol and Label Definition

Table 1-4 Symbol and Label Definition Table

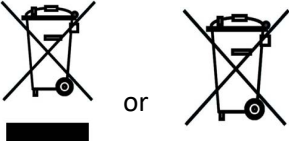
Symbol	Classification
	Manufacturer.
	Date of Manufacture.
	Consult Accompanying Documents.
	Consult Accompanying Documents.
	Serial Number.
	Part Number.
	Do Not Loop The Cable Warning Symbol.
	Do Not Loop The Cable Warning Symbol.
	Surface May Be Hot.
	This symbol indicates that the waste of electrical and electronic equipment must not be disposed as unsorted municipal waste and must be collected separately. Contact an authorized representative of the manufacturer for information concerning the decommissioning of your equipment.

Table 1-4 Symbol and Label Definition Table (Continued)

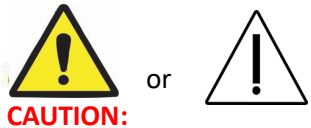
Symbol	Classification
	Receive Only Coil symbol. The Receive Only Symbol may be in color or black and white.
	Caution symbol. Indicates a hazard that may cause a minor injury or property damage if instructions are not followed.
	Warning symbol. Indicates a hazard that may cause severe personal injury, death, or substantial property damage if instructions are not followed.
	Do not insert with this side up.
	CE Label.
	Fragile - Do not drop or handle package roughly.
	Keep the packaging out of the rain. Keep packaging dry.
	Field Replaceable Unit part number.
	Revision of Part.
	Field Strength.

Table 1-4 Symbol and Label Definition Table (Continued)

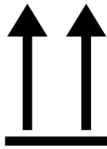
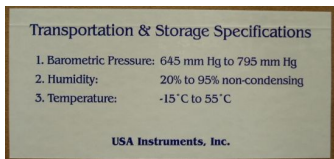
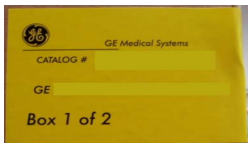
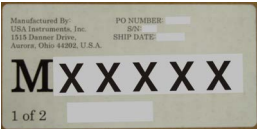
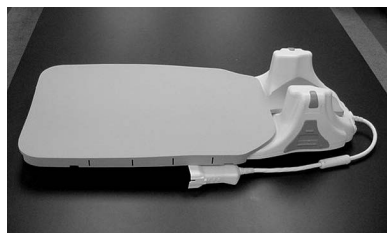
Symbol	Classification
	This end up. The arrows point to the side of the package that is to remain facing up.
	Keep packaging dry. Do not drop or handle package roughly. The arrow points to the side of the package that is to remain facing up.
	Temperature Limitation.
	Humidity Limitation.
	Atmospheric Pressure Limitation.
	Legacy Transportation and Storage Specifications Label.
	Shipping content label. Identifies the catalog number, GE HealthCare part number and the number of boxes in the shipment.
	Identifies contents, purchase order number, part number, quantity, shipping slip number, serial number and date shipped.

Table 1-4 Symbol and Label Definition Table (Continued)

Symbol	Classification
	Electronic Instructions for Use.
	Symbol indicates that this equipment generates, uses, and can radiate radio frequency energy.
	Symbol indicates that this product is a medical device.
	Swiss Authorized Representative.
	European Authorized Representative.
	Important indicates information where adherence to procedures is crucial or where your comprehension is necessary to apply a concept or effectively use the product.
	Shipping label. Identifies manufacturer, purchase order number, serial number, ship date, part numbers and number of boxes in the shipment.

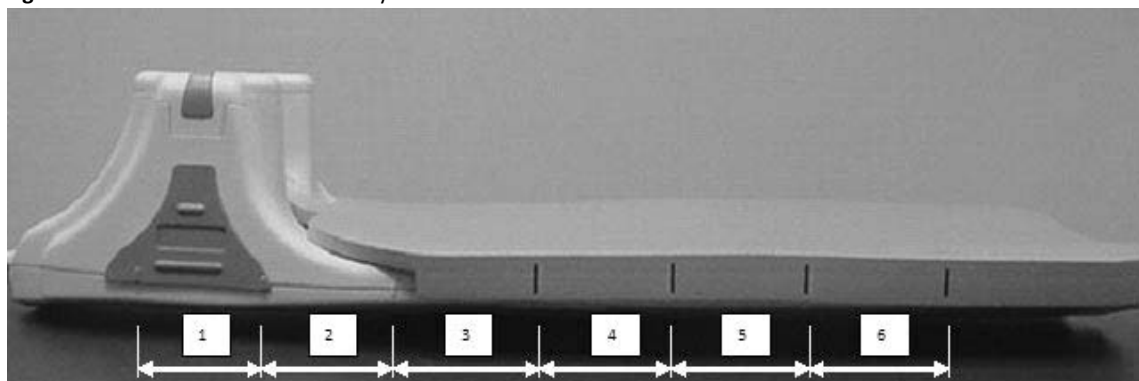
Coil Description



The 3.0T HD CTL Array is an 8-channel, receive-only coil designed to provide optimum signal-to-noise ratio and uniform coverage of the spine anatomy. It is specifically used for obtaining diagnostic images of the cervical-thoracic-lumbar regions.

The CTL coil consists of 12 elements arranged in six sections along the head-to-foot direction with 2 coil elements per section. Each section of the coil covers a different part of the patient's spine, allowing the operator to adjust for patient height and desired coil mode. In total, twelve coil modes are provided.

Figure 1-1 3.0T 8-Channel HD CTL Array



The 3.0T HD CTL Array is designed to be used by a Registered MRI Technologist. The Registered MRI Technologist will operate the scanner from the control room. If the patient is claustrophobic another MRI Technologist or clinical staff member may stay in the magnet room with the patient.

Coil Specifications

Table 1-5 3.0T HD CTL Array Specifications:

Field Strength	Model	Part #	Frequency
3.0T	Premier III Phased Array CTL Spine Coil (Receive Only)	2413107	127.73 MHz
3.0T	3.0T HD 8CH CTL Array	2415542	127.73 MHz

Coil Dimensions and Weight

NOTE: Store the coil in an area that exceeds the dimensions of the coil.

Dimensions

47 cm Wide x 113.5 cm Long x 23.6 cm High
(18.5 in Wide x 44.7 in Long x 9.3 in High)

Weight - 9.1 kg (20 lb)

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Chapter 2

Safety

Introduction to Safety

Patient safety and comfort are the primary concerns during the scanning procedure. Always follow proper safety, operating and maintenance procedures to ensure that the patient is not exposed to electrical or mechanical hazards that may cause injuries.

Route cables through the center of the magnet bore. Routing cable near the sides of the bore increases the likelihood of cable heating from induced currents. Place cables under the cushion whenever possible.

Keep the length of the cable in the bore to a minimum. When possible, avoid bending the cable 180 degrees. Route cables out of the bore in the most direct way possible, without looping or coiling.

Ensure that the patient is comfortably positioned.

Monitor the patient periodically. If the patient reports heating, burning or tingling sensations, **Stop the scan immediately.**

All personnel using this coil must be instructed in its proper use. Personnel must observe all warnings and cautions appearing in this manual.

Intended Use/Indications for Use for the 3.0T 8-Channel HD CTL Array

The 3.0T 8-Channel CTL Array Coil is designed to provide Magnetic Resonance Images of the cervical, thoracic, and lumbar regions of the spine. The 3.0T 8-Channel CTL Array Coil is designed for use with the GE HealthCare 3.0T scanner manufactured by GE HealthCare. The nucleus excited is hydrogen.

Contraindications

You should be aware of the contraindications for use related to the strong magnetic field of the MR system. Read this Safety chapter in this manual and then either the MR Safety Guide or the safety chapter in the operator manual provided with your system for contraindications. Follow screening instructions outlined in your MRI system documentation as well as the procedures established by your site. Use of this coil does not require additional MRI safety screening.

Patient Safety



Patient burn risk. Do not cross or loop coil cables.
Do not loop RF receive coil cables and ECG leads.



WARNING:



Do not allow the coil cable to come into contact with the skin of the patient. Use a thermal resistant material or pad to keep the cable from touching the patient. Failure to do so may result in patient burns.

WARNING:



Do not use accessories (e.g. pads or straps) that have not been specifically tested and approved for use in the MR environment. Use only GE HealthCare recommended accessories. Use of non-approved accessories may result in patient burns or injuries or image degradation. Even auxiliary devices labeled as compatible with MR equipment are capable of causing injury if the manufacturer's instructions are not followed.

WARNING:



RF can cause localized heating at contact points between adjacent body parts when a loop is formed. Such localized heating can result in a tingling sensation, discomfort, or burns. Loops may be formed by clasped hands, crossed arms, legs or feet, when a patient's hands are touching another body part, or when a female patient's breasts are compressed to her chest. Use pads between body parts to avoid creating a loop with adjacent body parts.

WARNING:



Always place appropriate non-conductive padding between the surface coil and the skin of the patient to prevent burn injuries.

WARNING:



WARNING:

Prior to patient placement in the coil, assure that any breached or compromised patient skin surfaces that come in contact with the coil have been adequately bandaged or covered.



CAUTION:

To avoid injuring the patient or damaging the coil, watch carefully as the table moves into the bore. Stop advancing the table if the patient or any part of the coil comes into contact with the bore.

Equipment Safety



WARNING:

Auxiliary equipment (e.g. gating equipment, vital signs monitoring systems and RF coils) that has not been specifically tested and approved may result in burns or other injuries to the patient. Such equipment may also interfere with proper operation of this coil. Use only equipment approved for the MR environment.



WARNING:

This coil is not suitable for use in the presence of flammable anesthetic mixture with air, oxygen or nitrous oxide.



WARNING:

Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any unused accessory device, coil, or cable from the magnet before using this coil.



WARNING:

Route cables through the center of the magnet bore. Place cables under the pads when possible. Routing cables near the sides of the bore increases the likelihood of cable heating (from induced currents).



CAUTION:

RF coupling risk. Looped cables may cause RF coupling and degrade the scan performance of the coil. Do not cross or loop coil cables.



Electrical and Mechanical Safety



WARNING:

Electric shock hazard. This coil consists of electrical and mechanical components. Tampering with the coil by untrained personnel can be hazardous to the patients and equipment. Only properly trained and qualified personnel should service the coil.



WARNING:

Use only coils, cables and accessories that are in good condition. Before using the coil, visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE HealthCare Service Engineer.



WARNING:

Electric shock hazard. No user serviceable parts. Refer service to qualified service personnel.

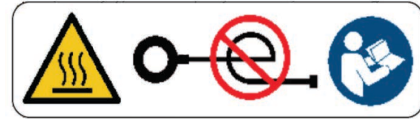
Definitions For Labels Located On The Coil

The labels shown below may be on the coil cable and/or cable housing assemblies.



WARNING:

The label shown to the right may be found on the coil cable. It contains three symbols: surface may be hot, do not loop, and consult accompanying documents.



WARNING:

The label shown to the right may be found on the coil cable. It contains two symbols: warning and do not loop.



WARNING:

Do not insert the connector plug with this side up.



The serial number label for the coil also contains several symbols. This label specifies date of manufacture, serial number, field replaceable unit number, field strength, place of manufacture, part number and revision. This label also has several symbols that are defined in this manual that will help the user better understand the specifics of the coil.

Modifications



WARNING:

Do not modify this equipment without authorization of GE HealthCare.

Any changes or modifications to this coil must be approved and performed by GE HealthCare prior to installation.

Emergency Procedures and Incident Reporting

If a patient requires emergency assistance:

- Stop the scan if one is in progress.
- Remove the patient from the scan room if medical treatment is needed.
- Please refer to MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part #2381696, if you received one with your system, for patient emergency procedures.

Reporting of Serious Incidents

Any serious incident related to the use of this GE HealthCare device should be reported to both the manufacturer and the health authority/competent authority where the device is installed.

To report to GE HealthCare:

- Either contact your local service representative
- Or report to: In-box.complaints@ge.com

Please provide the following information:

- The catalogue number or the model designation of the device as stated on its identification plate affixed on the device
- The SystemID/serial number/lot number of the device
- Date of incident
- Description of incident, including any patient or user impact/injury
- Your contact information (facility, address, contact name, title, and telephone number)

Reporting of Incidents with the Coil

In the unlikely event that a coil creates smoke, sparks or makes an unusually loud noise:

- Contact your local service representative to report the incident.
- Do not use the coil until a GE HealthCare Service Representative inspects the coil and indicates it is safe to use.

Users should immediately report to GE HealthCare any incidents or injuries that result from coil operation involving a patient, operator, or maintenance employee.

If an accident occurs as a result of coil operation, do not operate the equipment until an authorized investigation has been conducted.



CAUTION:

Users should place a service call any time the coil is dropped or mishandled. A GE HealthCare Service Representative should inspect the coil after it has been dropped or mishandled to ensure the coil is safe to use.

MR Safety Information

Please refer to MR Safety chapter in your MR system operator manual or refer to the MR Safety Guide, part #2381696, if you received one with your system. The MR Safety documentation describes the safety information you and the physicians must understand thoroughly before you begin to use the MR system. If you need additional training, seek assistance from qualified GE HealthCare personnel.

This equipment is intended for use by qualified personnel only.

This manual should be kept with the equipment and be readily available at all times. It is important for you to periodically review the procedures and safety precautions. **It is important for you to read and understand the contents of this manual before attempting to use this product.**

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Chapter 3

Coil and Patient Setup

Coil Installation

Coil installation is to be completed by a GE HealthCare Service Representative or a third-party service provider. To have the coil installed, please call your GE HealthCare Service Representative or your third-party service provider. Service representatives have access to documents that contain instructions for safely and compliantly installing, testing, and maintaining your MR Coil.

Lifting and Transporting the Coil

- Pick up the coil by firmly gripping it on both sides.
- Do not carry the coil by the cable.



Do not carry any of the coil components by the cable. Damage to the coil component may occur. The coil may not work if damaged.

CAUTION:

Coil Mode Configurations

To optimize image quality using the CTL coil, select a coil mode appropriate to the anatomy of interest and the desired FOV. Improper use of the coil is the major cause of image artifacts.

Table 3-1 Modes of Coil Operation :

3.0T HD CTL Array Modes	3.0T HD CTL Array, A/B Port Modes	Clinical Use	Selected Sections	S/I Coverage	Figure Reference
8USCTLTOP	HDCTL TOP	Cervical and thoracic spine	1, 2, 3, 4	50 cm	3-1
8USCTLMID	HDCTL MID	Thoracic and lumbar spine	2, 3, 4, 5	50 cm	3-2
8USCTLBOT	HDCTL BOT	Thoracic and lumbar spine	3, 4, 5, 6	50 cm	3-3
8USCTL123	HDCTL 123	Cervical and thoracic spine	1, 2, 3	38 cm	3-4
8USCTL234	HDCTL 234	Thoracic spine	2, 3, 4	38 cm	3-5
8USCTL345	HDCTL 345	Thoracic and lumbar spine	3, 4, 5	38 cm	3-6
8USCTL456	HDCTL 456	Thoracic and lumbar spine	4, 5, 6	38 cm	3-7
8USCTL12	HDCTL 12	Cervical spine	1, 2	26 cm	3-8
8USCTL23	HDCTL 23	Thoracic spine	2, 3	26 cm	3-9
8USCTL34	HDCTL 34	Thoracic spine	3, 4	26 cm	3-10
8USCTL45	HDCTL 45	Thoracic and lumbar spine	4, 5	26 cm	3-11
8USCTL56	HDCTL 56	Lumbar spine	5, 6	26 cm	3-12

These modes of operation can be enabled at the system user interface.

Figure 3-1 Eight coil elements are active for this mode:

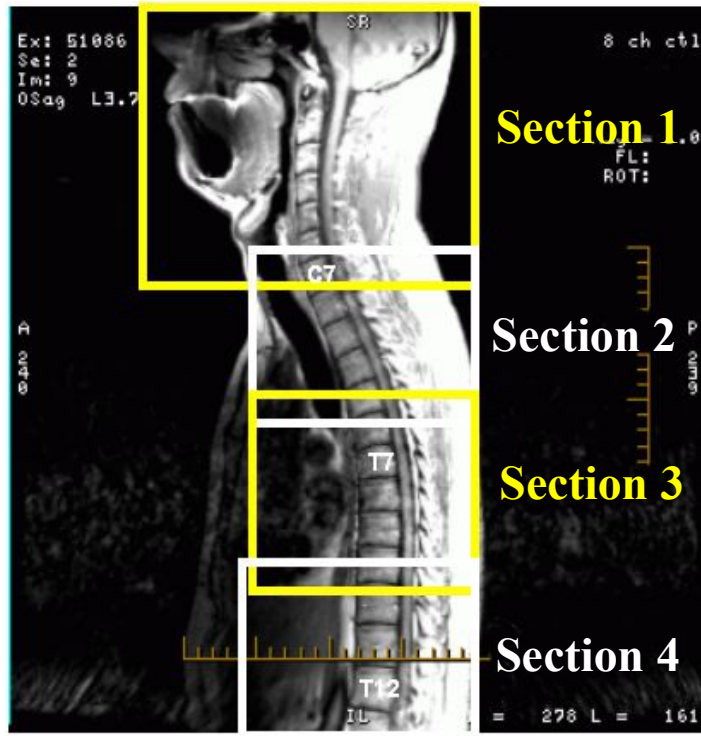


Figure 3-2 Eight coil elements are active for this mode:

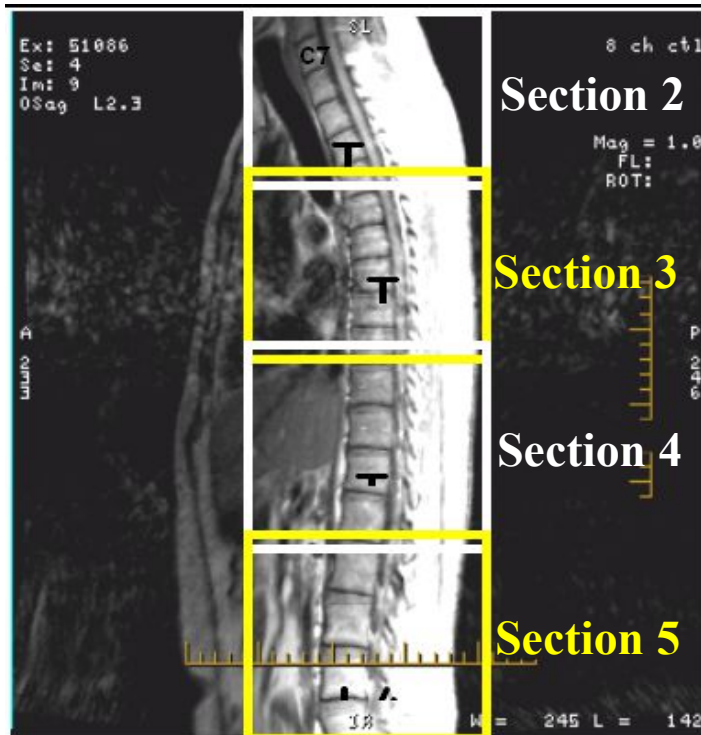


Figure 3-3 Eight coil elements are active for this mode:

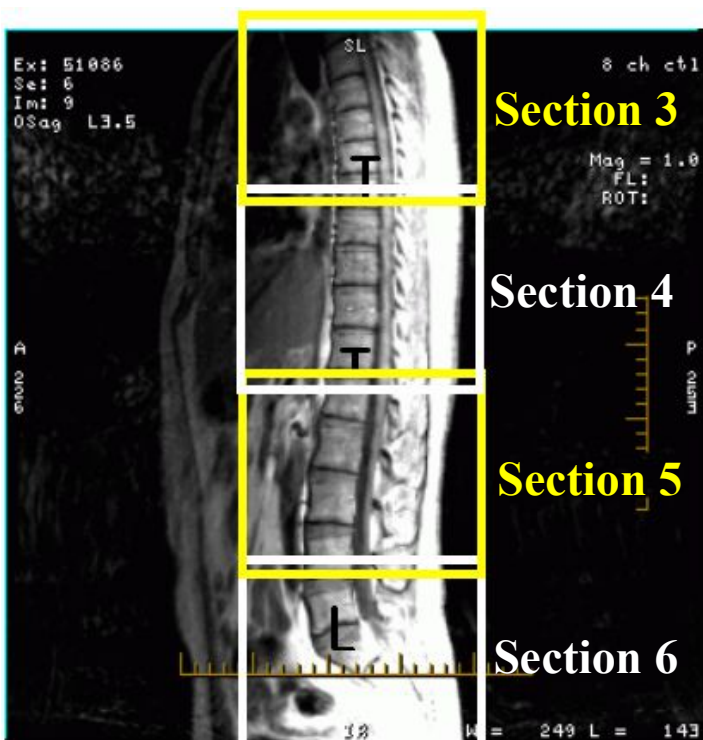


Figure 3-4 Six coil elements are active for this mode:

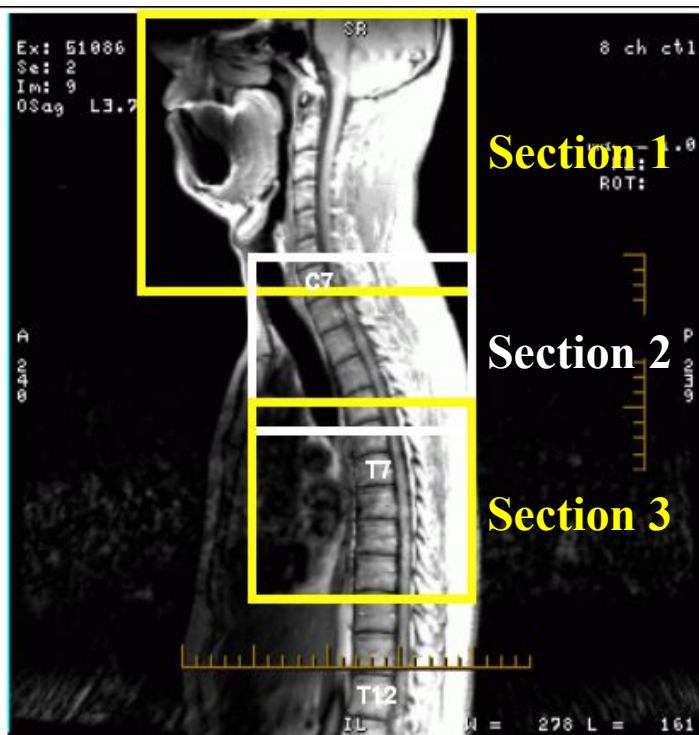


Figure 3-5 Six coil elements are active for this mode:

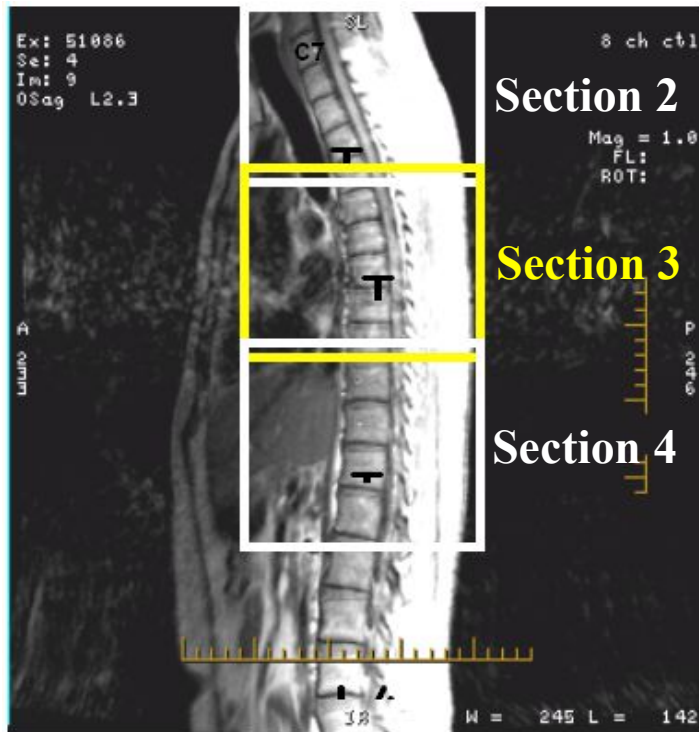


Figure 3-6 Six coil elements are active for this mode:

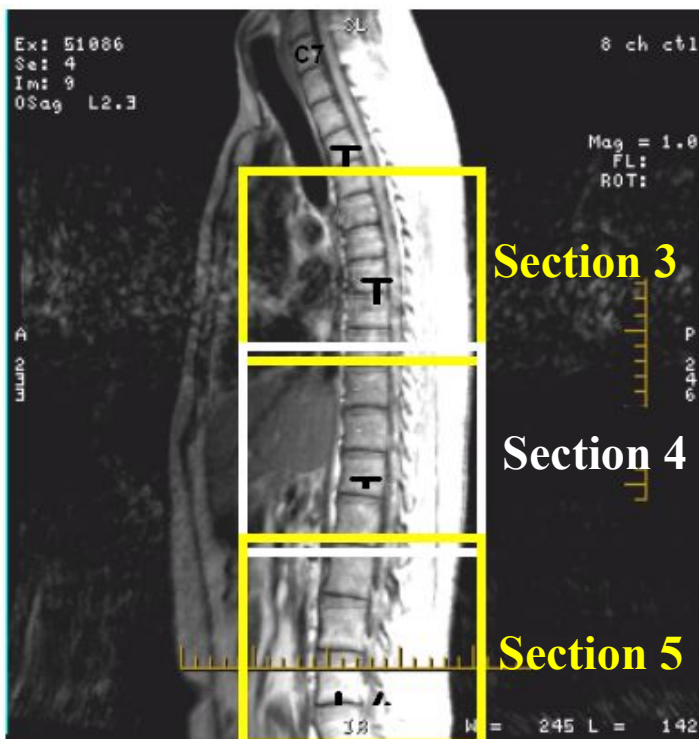


Figure 3-7 Six coil elements are active for this mode:



Figure 3-8 Four coil elements are active for this mode:

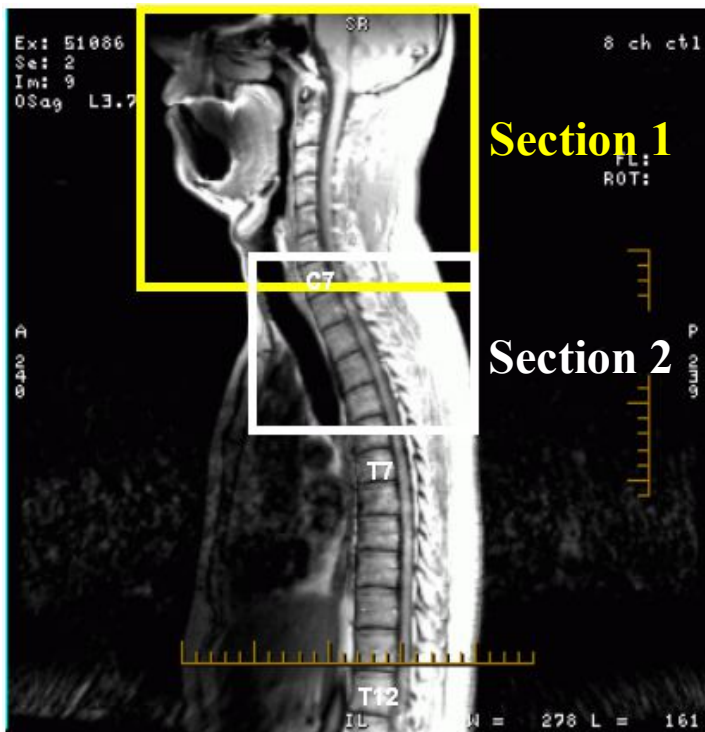


Figure 3-9 Four coil elements are active for this mode:

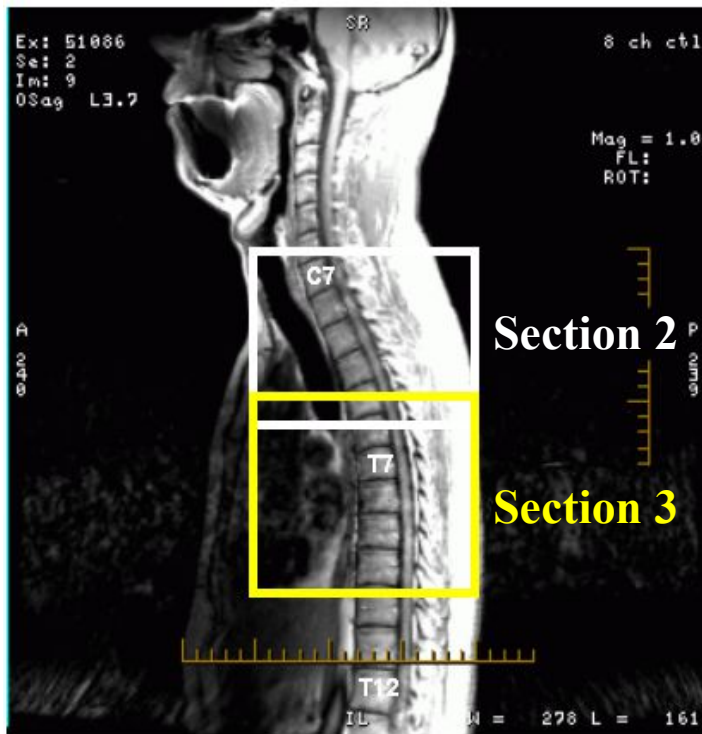


Figure 3-10 Four coil elements are active for this mode:

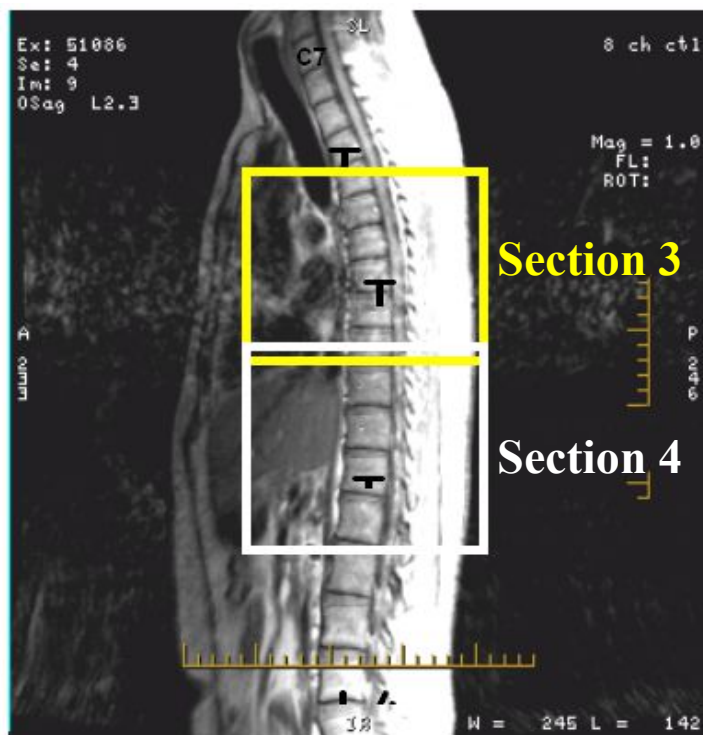


Figure 3-11 Four coil elements are active for this mode:

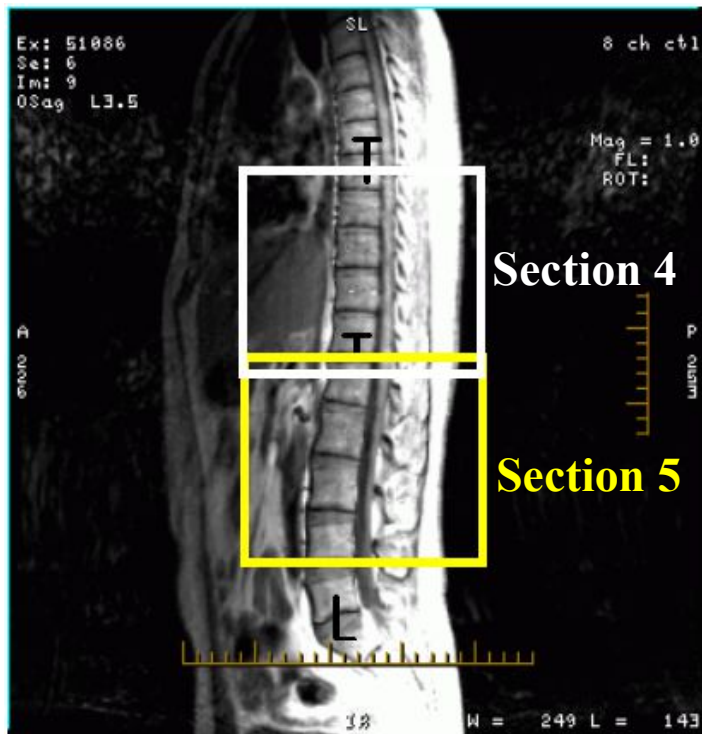


Figure 3-12 Four coil elements are active for this mode:



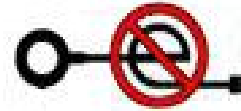
Positioning the Coil and Patient


WARNING:

RF coupling danger. Patient burns may result from RF coupling of this device with other devices remaining in the magnet. Remove any other coil or unused accessory device from the magnet before using this coil.


WARNING:

Patient burn risk. Do not cross or loop coil cables. Do not allow any cable to touch the patient. Use a thermal-resistant material or pad to prevent cables from touching the patient.


CAUTION:

Be sure the coil is unplugged before lowering the table. Failure to do so may damage the coil, cable or connector. The coil may not work if damaged.

Table 3-2 Positioning the Coil and Patient on the Scan Table:

Visual	Description
	1. Position the coil (with comfort pad) on the table near the bore of the magnet (left).
	2. Position the patient head first, supine, with the top of the shoulders resting firmly against the coil sides. Use earpads to secure the patient's head in the coil. Position the patient's arms to his or her sides. NOTE: A cotton sheet should be used as necessary to keep the patient's skin from touching the coil.

Table 3-2 Positioning the Coil and Patient on the Scan Table: (Continued)

Visual	Description
 3.0T Legacy Connector  HD Connector	3. Plug the coil cable into the appropriately labeled port of the system receptacle. NOTE: The coil connector and cable should always be routed directly to the magnet and should not form loops. 4. Localize the anatomy and advance to scan.

Localize Anatomy and Choose Mode Configuration

Align the laser light on the region of interest according to the mode chosen. Refer to **Chapter 3, page 3 - 2 - Coil Mode Configurations**, for detailed information.

Recommended Scan Protocols

Use the GE HealthCare recommended protocols for scanning with this coil for optimum results and best performance.

Specific choice of protocols is entirely the responsibility of the radiologist and/or technologist.

Chapter 4

Coil Maintenance

Coil Maintenance - Introduction

To maintain the highest possible image quality and to protect patients and personnel from potential injury or illness, it is important to keep the coil clean and well maintained. This chapter lists basic methods for keeping the coil safe and sanitary. Also included is a list of accessories that will enhance the utility of your coil. Refer to the contact information for questions or comments.

Quality Assurance

Coil Quality Assurance tests must be run by a GE HealthCare Service Representative or a third party service provider. To have a Quality Assurance test run on a coil, please call your GE HealthCare Service Representative or your third party service provider.

If you do not have a GE HealthCare Service Representative or third party service provider, use the Class A Service CD that was shipped with your MR System. This CD contains information on phantom setup and the usage of the MCQA tool. Please contact GE HealthCare with any questions or concerns.

Coil Care

The coil must be inspected before each use. Visually check each coil component, cable and accessory to ensure that there is no external damage. If any coil component, cable or accessory is suspected of not being in good condition, discontinue use and contact a GE HealthCare Service Engineer.

- The coil is damaged in a way that exposed internal conductors or foam.
- The coil becomes severely soiled beyond what can be cleaned according to the instructions found later in this chapter.
- There are image artifacts seen during scanning.

All FRU's (Field Replaceable Units) must be replaced by a GE HealthCare Service Engineer or a third party service provider. Component part lists with descriptions, instructions for the safe replacement of components, circuit diagrams, and calibration and preventive maintenance instructions are available to service personnel upon request.

General Cleaning and Disinfection



WARNING:

Electric shock may occur if the coil is attached to the system during cleaning or when it is still wet. Detach coil connector from the scanner before attempting to clean the coil. Do not touch connectors with bare fingers. Never press sharp objects against connector surface. Do not reattach connector after cleaning until the coil has dried completely.



CAUTION:

Inspect pads for peeling or cracking. To prevent a biohazard, replace cracked or peeling pads before using.



CAUTION:

The coil contains sensitive electronic components that may be damaged by liquids. Do not spray or pour cleaning solution directly onto the coil. Do not submerge the coil in any solution. Under no circumstances should the coil be placed into any type of sterilizer.



CAUTION:

To avoid possible damage to equipment, do not use solutions containing amines, strong alkalis, quaternary ammonium chloride compounds, esters, iodine, aromatic or chlorinated hydrocarbons, or ketones. Do not use autoclaves or the industrial washers and dryers found in most hospitals or professional laundry services.

Clean and disinfect the Coil after removing from packaging prior to first use.

Then, cleaning and disinfection of patient contact surfaces is recommended following each use.

Non-patient contact surface cleaning and disinfection should be conducted following internal housekeeping procedures unless otherwise indicated in the following instructions.

Cleaning and Disinfection Recommendations for the MR System, Coils and Most Accessories

- **Clean** with commercially available wipes that contain 0.525% minimum sodium hypochlorite as the only active ingredient. If commercially available wipes are not available, then follow one of these instructions.
 - Clean with a lint free cloth saturated with 1:10 dilution of commonly available bleach containing a recommended minimum sodium hypochlorite of 5.25%. Dilute the bleach with tap water.
 - Use a lint free cloth that has been saturated with a 70% isopropyl alcohol solution made from 100% isopropyl alcohol and 30% tap water.
 - Use a lint free cloth with a commercially prepared solution of 70% concentration isopropyl alcohol.
 - Carefully clean all surfaces including cracks and crevices. Consider placing a wipe around a rigid implement (GE HealthCare Part Number 5878912 or equivalent) and use this to get into cracks. Use swabs covered with wipes or diluted cleaning agents to access blind holes and recessed areas.
 - Regardless of the method used, inspect to ensure visual cleanliness prior to disinfection. Repeat the cleaning process until all visible soil has been removed.

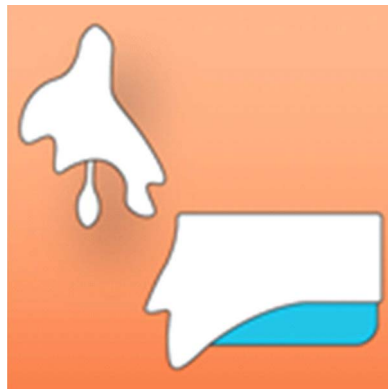


Figure 4-1 : Suggestions for rigid cleaning implements. The upper left image is a swab covered with a wipe. The lower right image is of a card covered with a wipe.



Figure 4-2 : Part number 5878912, example of a cleaning tool.

- **Disinfect** with commercially available wipes that contain 0.525% minimum sodium hypochlorite as the only active ingredient following the manufacturer's instructions. If commercially available wipes are not available, then follow these instructions.
 - Disinfect with a lint free cloth and a 1:10 dilution of commonly available bleach containing a recommended minimum sodium hypochlorite of 5.25%. Dilute the bleach with tap water.
 - For general disinfection or disinfection following cleaning of blood and/or body fluids, 5 minutes contact time is recommended.
 - Use the implements described above for disinfection, if necessary.
 - Refer to internal procedures or refer to publications such as "CDC Guideline for Disinfection and Sterilization in Healthcare Facilities," 2008 or latest revision, for guidance. Disinfectant may need to be reapplied to ensure surfaces remain wet for the duration of the selected contact time.
 - After you have cleaned and disinfected with bleach (sodium hypochlorite), wipe surfaces with a disposable lint free wipe that has been dampened with purified water to remove any remaining bleach residue and allow device to air dry indoors before use.
- Should your coil need to be returned to GE HealthCare for service, please clean and disinfect the component being returned as instructed above.
- Dispose of any materials used to clean the coil and the pads according to all federal, state, and local regulations.

Accessories and Replaceable Components

Coil Replacement parts (FRU's - Field Replaceable Units) should be ordered by a GE HealthCare Service Representative or a third party service provider.

If you do not have a GE HealthCare Service Representative or third party service provider, please contact GE HealthCare.

Table 4-1 3.0T CTL Array Accessories and Replaceable Components:

Part Name	Part Number
Wedge/Ear Pad Set	2414538
Patient Comfort Pad	2413107-5
Disposable Ear Pad Covers	2380634-8

Glossary

C	Celsius
CDC	Centers for Disease Control and Prevention - the national health agency of the United States
cm	Centimeter
ECG	Electrocardiogram
eIFU	Electronic Instructions for Use
EMC	Electromagnetic Compatibility
F	Fahrenheit
hPa	A unit of pressure; Hecto Pascal
in	Inch
IP Rating	Ingress Protection rating. The effectiveness of the seal of the coil against intrusion from foreign bodies and moisture
kg	Kilogram
lbs	Pounds
MCQA	Multicoil Quality Assurance
MHz	Megahertz
mm	Millimeters
mm Hg	A unit of pressure; millimeters of mercury

MRI	Magnetic Resonance Imaging - The electromagnetic signal (in the radio frequency range) produced by the precession of the transverse magnetization of the spins. The rotation of the transverse magnetization induces a voltage in the coil. This voltage is amplified by the receiver.
QA	Quality Assurance
RF	Radio Frequency - The frequency (intermediate between audio and infrared frequencies) used in magnetic resonance systems to excite nuclei to resonance.
Type BF	The rating of protection against electrical shock for the coil